

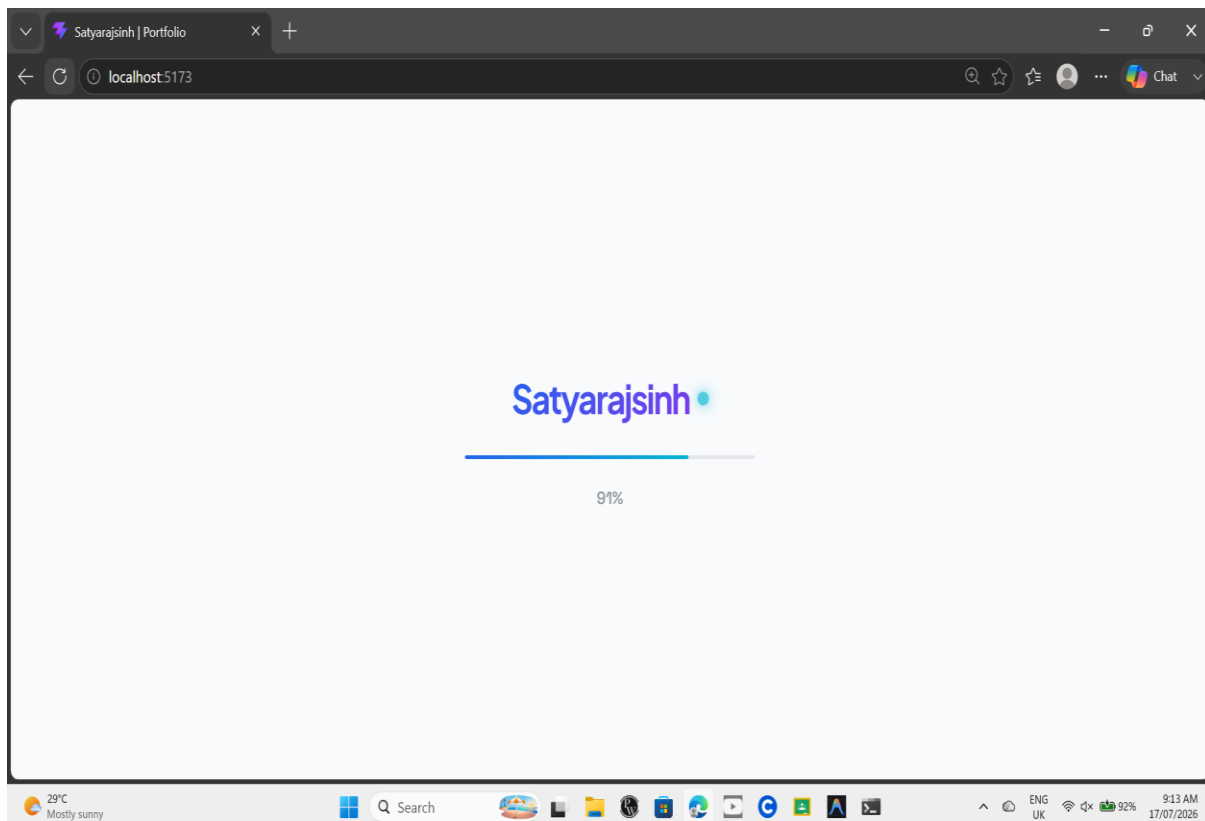
PRACTICAL-1

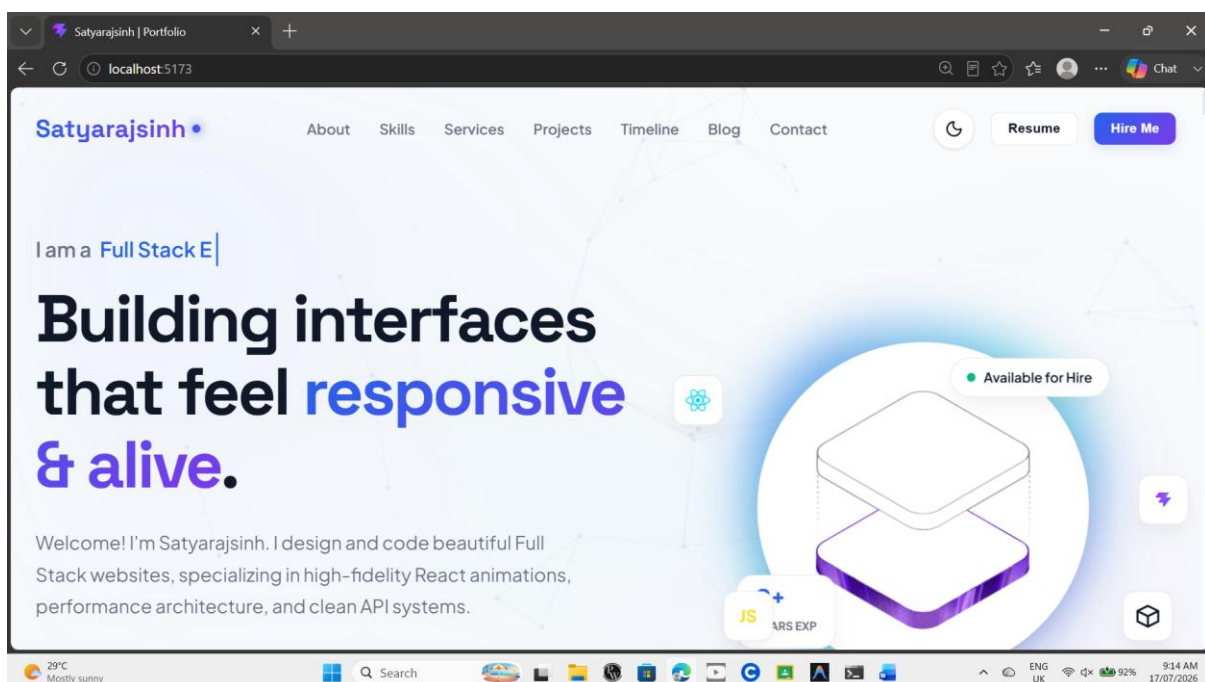
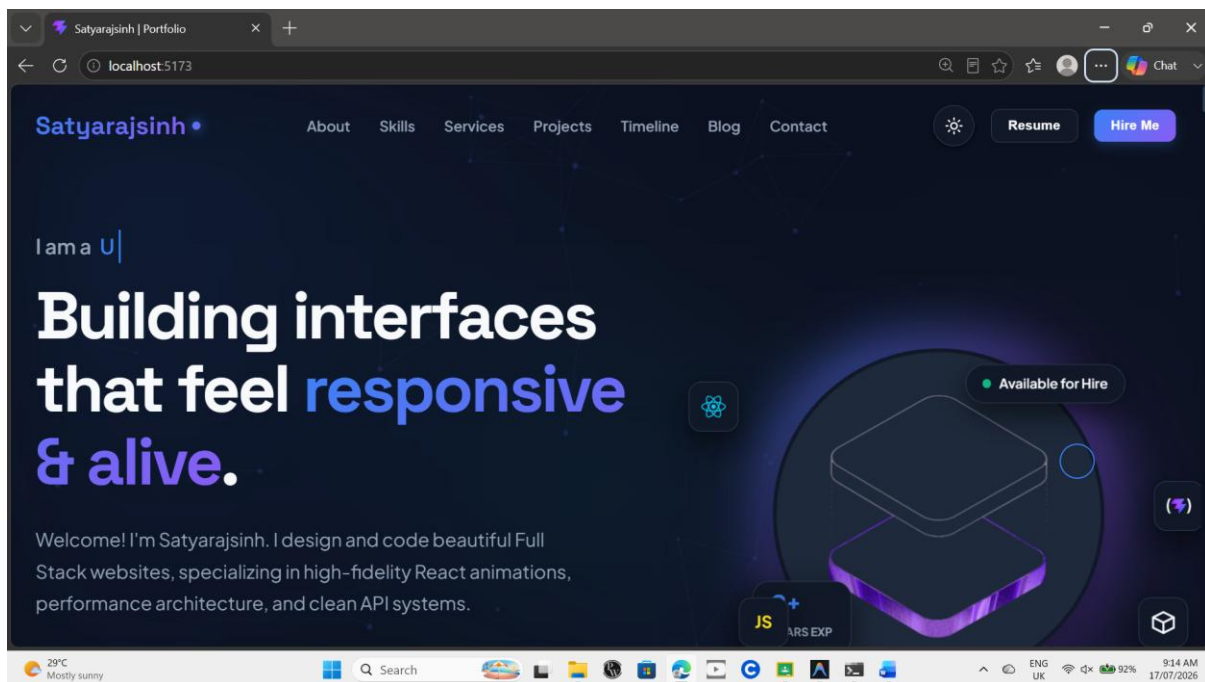
Problem Defination:

- Create a React application using Vite for a student portfolio page.
- The application must include at least 4 reusable components: Header,About, Skills, and Footer.
- Each component must be independently structured and composed into a single-page layout.
- No logic or JSX should be duplicated across components.
- Props must be used to pass at least one piece of data into a minimum of 2 components.

GitHub: https://github.com/24dit061-Satyarajsinh/24DIT061_ITUE301-AWD

OUTPUT:





The screenshot shows a web browser window with the address bar displaying 'localhost:5173'. The page is titled 'ABOUT ME' and 'My Journey & Background'. On the left, there is a dark blue box containing a JSON object named 'developer_details.json' with the following content:

```
const developer = {
  name: "Satyarajsinh",
  nationality: "Indian",
  status: "Active Freelancing",
  currentFocus: "React & Full Stack APIs",
  hoursCoding: 10000+,
  remoteFriendly: true
}
```

Below the JSON box, there are two sections: 'Quick Facts' showing 'Location: India' and 'Interests' showing 'Coding' and 'UI Design'. A 'Download Full CV' button is also present. On the right, the text reads: 'Engineering digital experiences with precision'. Below this, it states: 'I am a results-oriented Software Developer who finds joy in solving complex logical problems. With a core focus on frontend engineering (React/Vite) and backend integrations, I convert business ideas into clean, maintainable web products.' and 'I am highly dedicated to optimizing client load times, applying clean visual spacing rules, and structuring clean visual designs that feel fluid, responsive, and world-class.'

The browser's taskbar at the bottom shows the system clock as 9:15 AM on 17/07/2026, with a battery level of 92% and network status.

The screenshot shows a web browser window with the address bar displaying 'localhost:5173'. The page is divided into two main sections: 'Work Experience' and 'Education'.

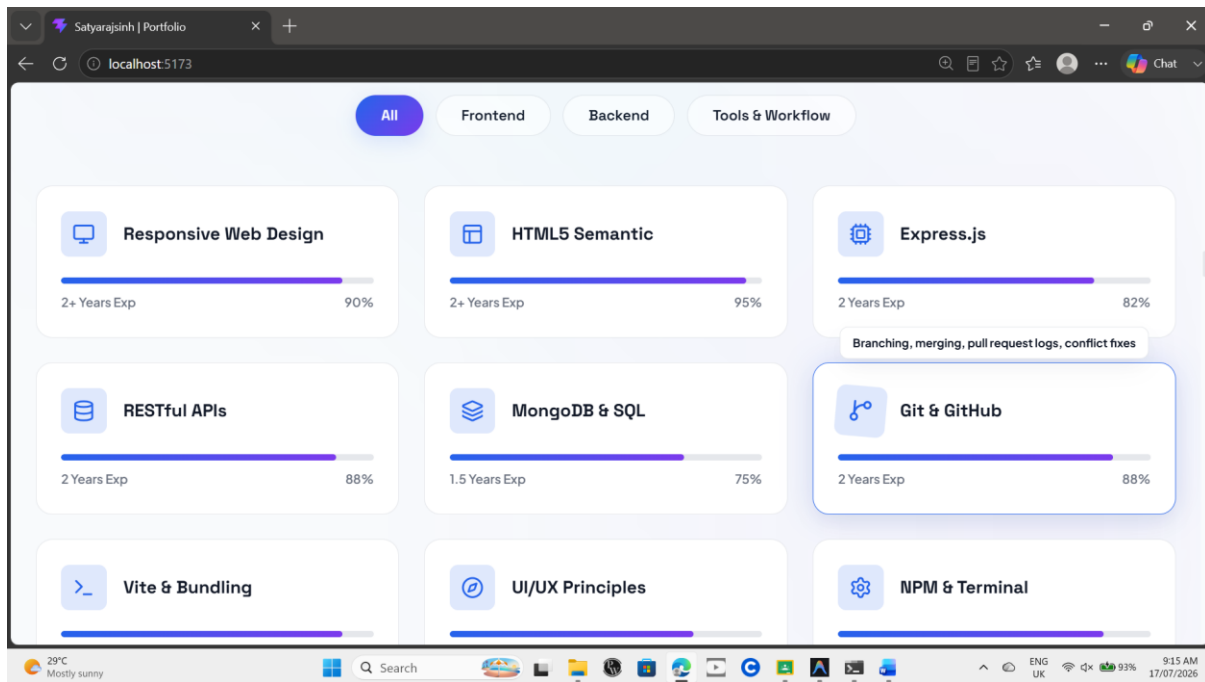
Work Experience:

- 2024 - Present**
Freelance Full-Stack Developer
Self-employed | Remote
Design and code premium front-end components, landing pages, and API servers. Collaborated directly with business owners to optimize SEO rankings and interface responsive fluid animations.
- 2023 - 2024**
Frontend Intern / Developer
Tech Solutions Inc.
Learned production developer pipelines. Built modular HTML/CSS/React widgets, resolved responsive grid layout failures, and structured REST integration layers.

Education:

- 2020 - 2024**
B.E. in Computer Engineering
GTU Technical University
Focused on software algorithms, databases, web technologies, and systems engineering. Graduated with honors, scoring a cumulative 8.5 CGPA.
- 2018 - 2020**
Higher Secondary Certificate
Science Stream | High School
Learned fundamental physics, mathematics, and programming logic foundations, graduating with a 90% grade average.

The browser's taskbar at the bottom shows the system clock as 9:15 AM on 17/07/2026, with a battery level of 93% and network status.



Key-Questions:

- Question:** What is the role of each component in the overall UI structure?

Answer: Each React component represents a specific part of the user interface, such as the Header, About, Skills, or Footer. Components help organize the application into small, independent, and reusable sections. This modular structure improves code readability, simplifies maintenance, and makes the application easier to scale and update.
- Question:** How does React re-render when props change?

Answer: When a parent component passes new props to a child component, React automatically re-renders the child component. It first compares the new Virtual DOM with the previous Virtual DOM to detect changes. Then, React updates only the affected elements in the real DOM, making the rendering process efficient and improving application performance.
- Question:** Why is component reusability important in large-scale applications?

Answer: Component reusability allows developers to build a component once and use it multiple times throughout the application. This reduces code duplication, ensures a consistent user interface, and speeds up development. It also makes debugging, testing, and maintaining large-scale applications much easier because changes only need to be made in one place.

Conclusion: This practical demonstrated how to build a React application using Vite with a reusable component-based architecture. Separate components such as Header, About, Skills, and Footer were created and combined into a single-page portfolio. The use of props enabled data sharing between components, resulting in a modular, maintainable, and efficient React application.